How to use Snowflake SQL from VS Code?

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Disclaimers:

- made with/for PC and Windows 11
- all opinions (and mistakes) are mine

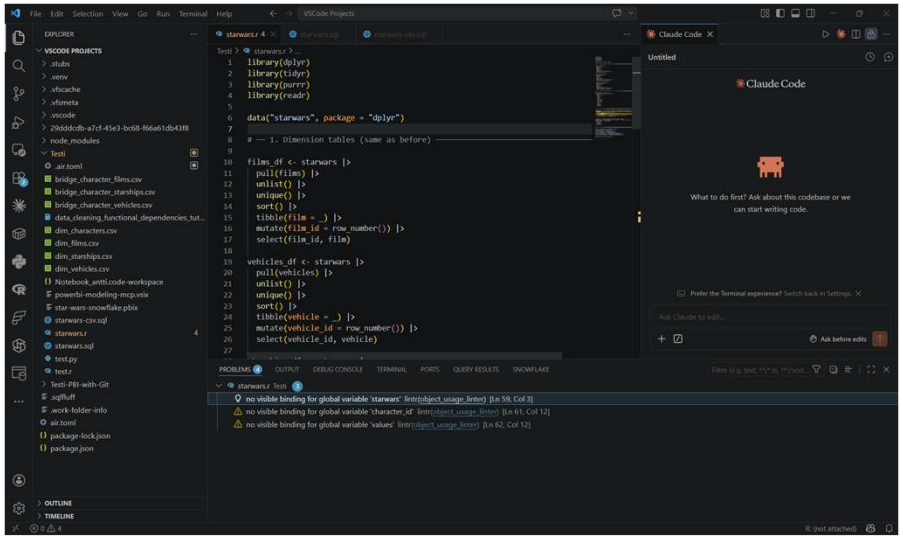
VSCode vs. Snowsight

VSCode vs. Snowsight

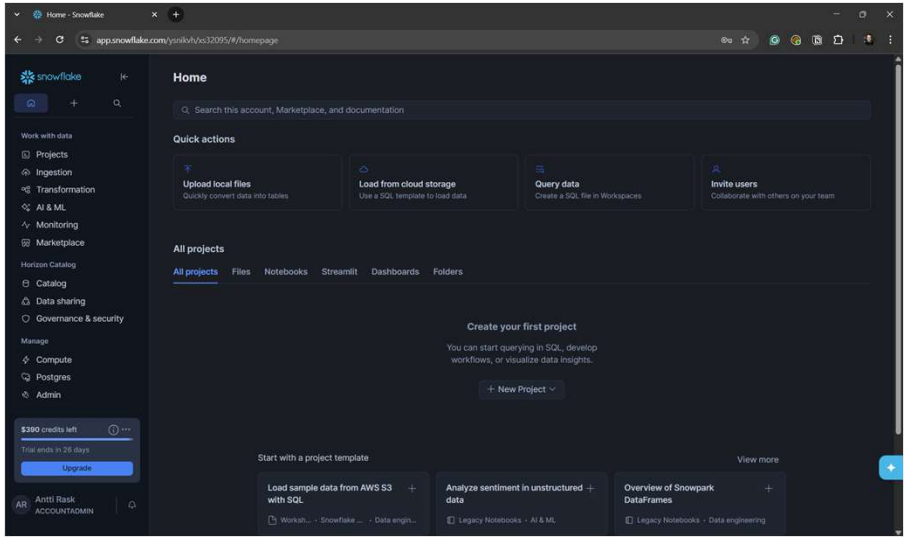
- This presentation was first made a year ago and was called Why use Snowflake SQL from VS Code (instead of Snowsight)?
- A lot has changed since that, mainly the Workspaces feature in Snowsight, which bridged the gap significantly
- Now it’s more about personal preference, but there are still some differences between the two features-wise

VSCode vs. Snowsight

VSCode (a.k.a. Visual Studio Code)



Snowsight



Editing

Feature	VSCode	Snowsight
SQL linting & formatting	Full control with **sqlfluff**, for instance	Built-in Format SQL (Workspaces only, limited style rules)
Regex search & replace	Multi-file (with Ctrl + Shift + F)	Within a single file

File & Project Management

Feature	VSCode	Snowsight
Queries as .sql files	Native, any file system structure	File-based in Workspaces, but stored in Snowflake's system
Folder organisation	Full OS file system, unlimited nesting	Folders supported in Workspaces
Share queries with team	Via Git / any file sharing	Shared workspaces with role-based access, within Snowflake org only
Works offline	Yes, edit without a connection (of course running a query requires a connection)	Browser-only, requires a connection

Results & Comparison

Feature	VSCode	Snowsight
Open results in new tab	Yes	Results stay in same worksheet pane
Export results	Direct to file system	Download CSV/JSON from UI
Column stats / data profiling	Not built in	Inline histograms, sum/avg per column
Query profile	Not available	Visual execution plan with node-by-node breakdown

Workflow & History

Feature	VSCode	Snowsight
Extensions / plugins	Vast ecosystem: linters, dbt, AI, any database	No extension model, but deep native integrations (dbt, Snowpark, Streamlit, Cortex AI)
AI coding assistant	GitHub Copilot, Cursor, Continue, etc.	Snowflake Copilot (native, schema-aware, improving)
Keyboard-first workflow	Command palette, full shortcut customisation	Improved with Monaco (open-sourced VSCode editor engine), mouse still needed more often

The one big thing that is still only available in Snowsight: Query Profile

What if I want to see the Query Profile?

```
Execute
SET character_name = 'Luke Skywalker';

Execute
WITH film_list AS (
    SELECT
        bf.character_id,
        ARRAY_AGG(f.film) AS films
    FROM starwars.public.bridge_character_films bf
    INNER JOIN starwars.public.dim_films f ON bf.film_id = f.film_id
    GROUP BY bf.character_id
),

vehicle_list AS (
    SELECT
        bv.character_id,
        ARRAY_AGG(v.vehicle) AS vehicle
    FROM starwars.public.bridge_charact
    INNER JOIN starwars.public.dim_vehi
    GROUP BY bv.character_id
),

starship_list AS (
    SELECT
        bs.character_id,
        ARRAY_AGG(s.starship) AS starships
```

1. Get the Query ID

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | QUERY RESULTS | SNOWFLAKE

QUERY RESULTS

Table | Chart

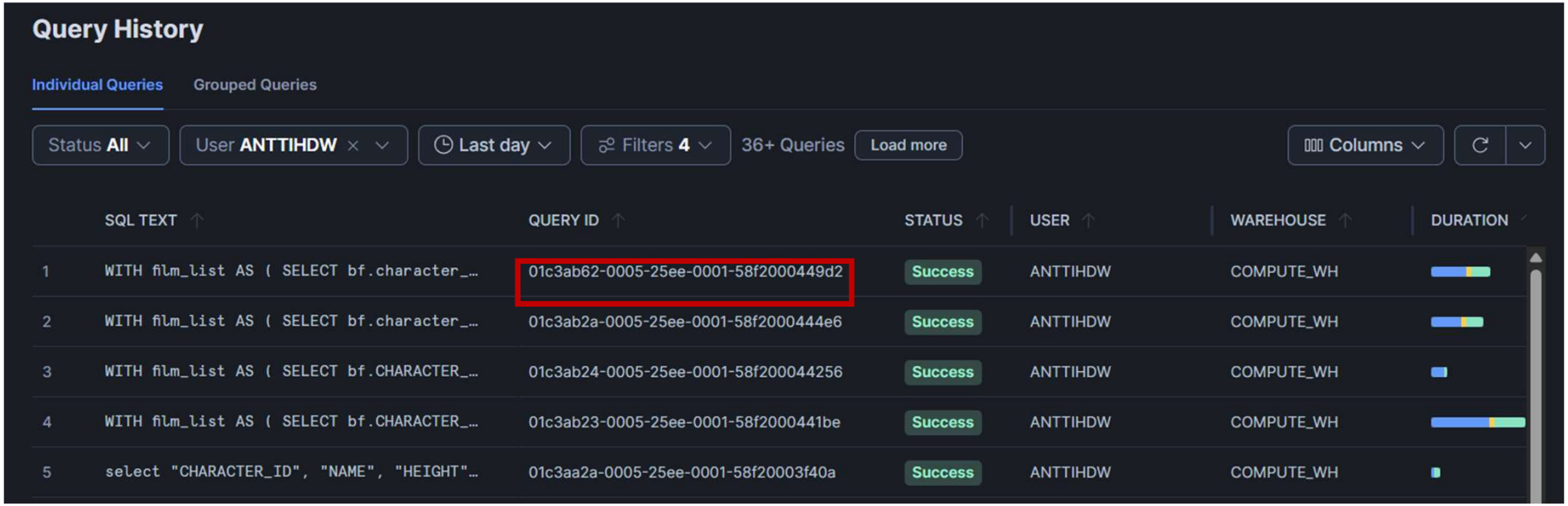
	NAME	HEIGHT	MASS	HAIR_COLOR	SKIN_COLOR	EYE_
1	Luke Skywalker	172	77	blond	fair	blue

Query Details

01c3ab62-0005-25ee-0001-58f2000449d2

Query Duration	857ms
Rows	1
End time	Mon Apr 13 2026
Scanned	11.5KB (100%)
Role	ACCOUNTADMIN
Warehouse	COMPUTE_WH

What if I want to see the Query Profile?

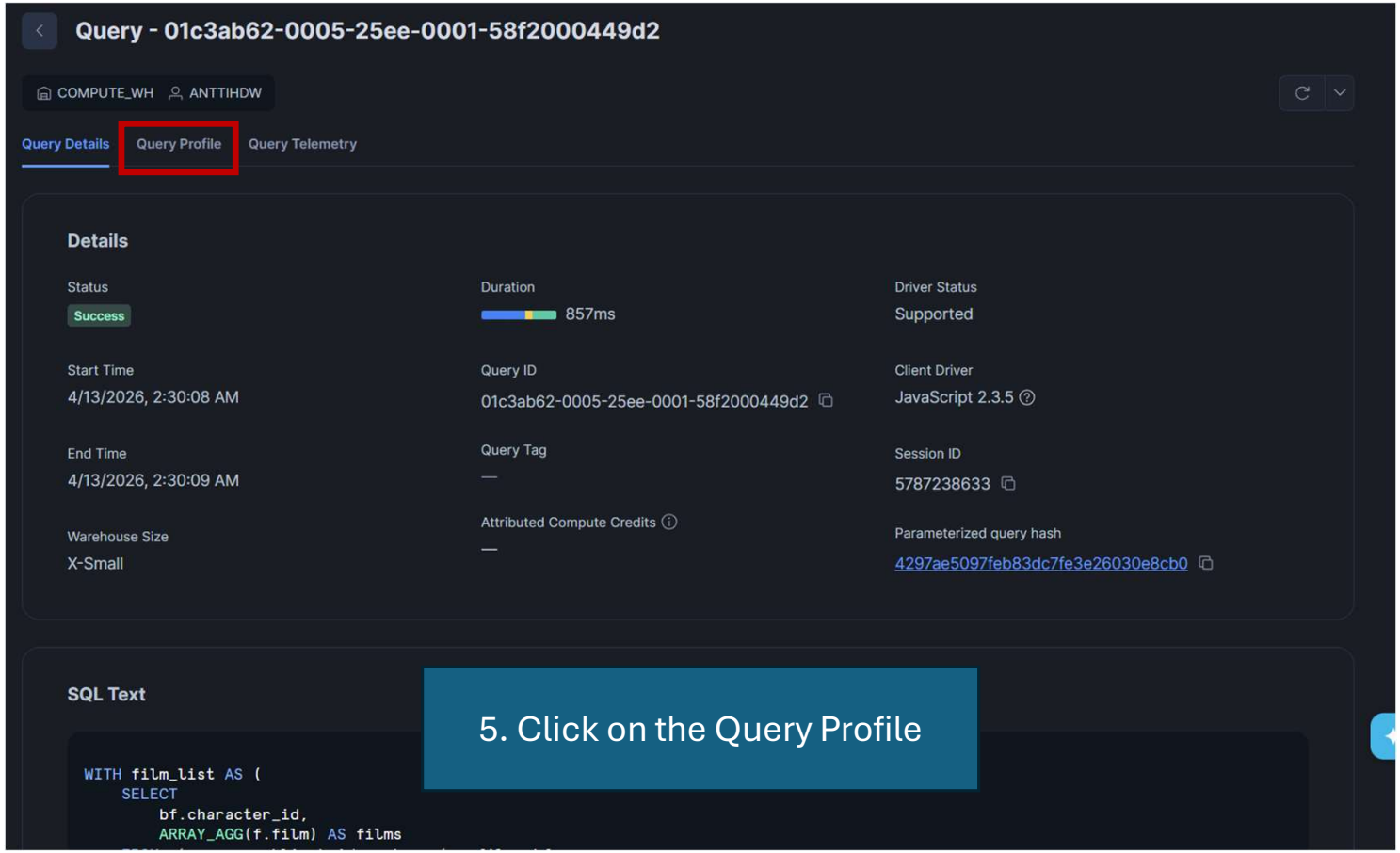


2. Open Snowsight

3. Find the query with a matching Query ID

4. Click on the Query ID

What if I want to see the Query Profile?



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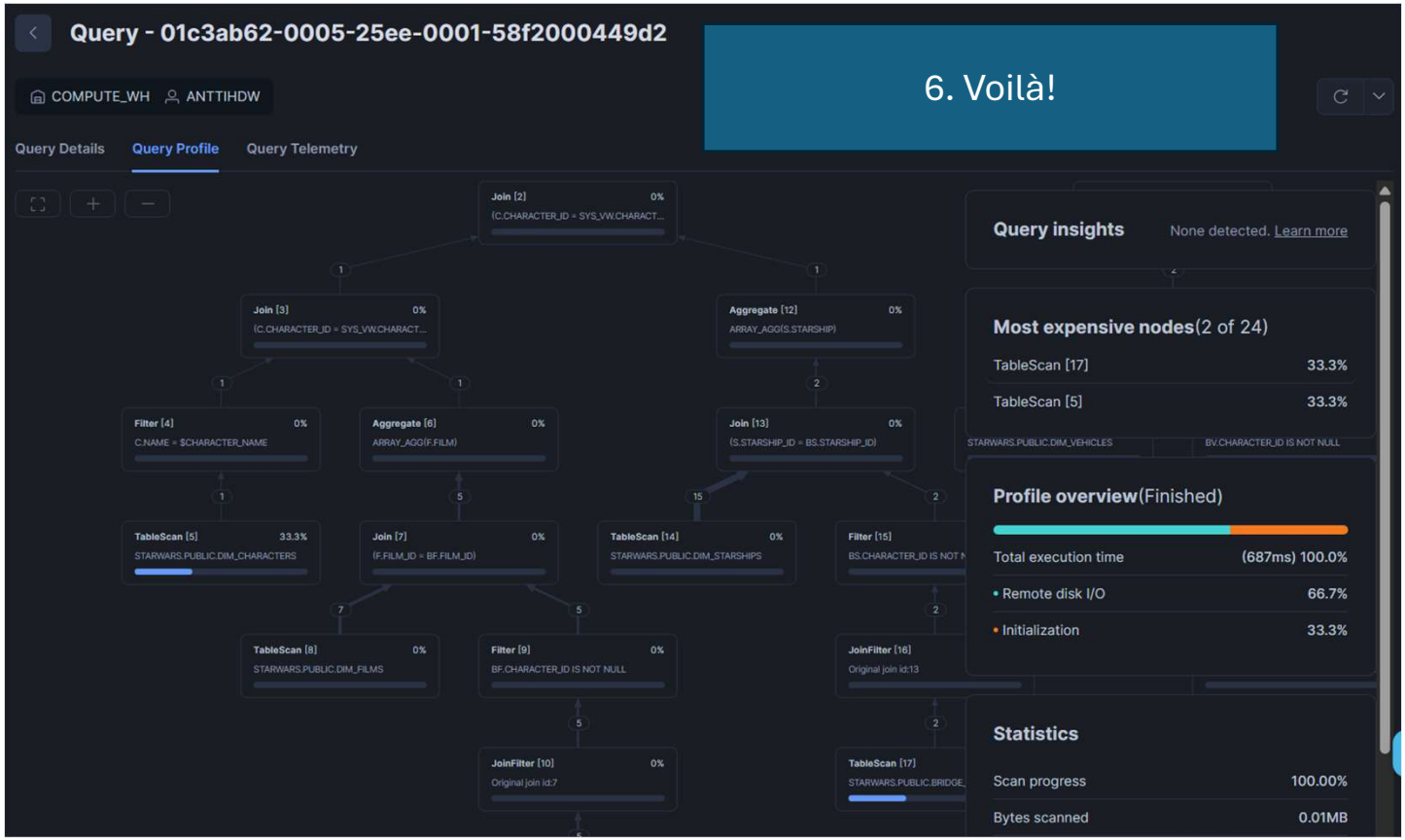

How to start using Snowflake SQL inside VS Code?

Setting up

1. Install Visual Studio Code (VSCode)
2. Open VSCode
3. Open Extensions (Ctrl + Shift + X) and install the following:
   - Snowflake
   - Snowflake VSCode
     - Not mandatory, but “will highlight keywords and functions for the Snowflake flavor of SQL in VS Code”
   - sqlfluff
     - Not mandatory, but it’s a “linter and auto-formatter for SQLfluff, a popular linting tool for SQL and dbt”
     - You need to also install SQLfluff (with Python)

Connecting to Snowflake

1. Click the Snowflake Icon on the activity bar
2. From the Account dropdown
   - Add Account Identifier
     - Run SELECT CURRENT_ORGANIZATION_NAME() || '-’ || CURRENT_ACCOUNT_NAME(); in Snowsight to get the ID
   - Auth Method: Depends on your account/organization
     - I’m using Username/Password with this Free trial account, but for my work account I used Single Sign-On

Demo time!

ggplot2 extended book (in progress)
https://ggplot2-extended-book.com/

LinkedIn
https://www.linkedin.com/in/anttirask/

Portfolio (apps, etc.)
https://youcanbeapirate.com/

Substack
https://youcanbeapirate.substack.com/

Want to become a colleague?
https://careers.nortal.com/

This presentation on GitHub: